dynamic attributes like current location or work schedule, which may have to be accessed from other systems outside the identity system.

### 2.2.4 The privacy problem

The pervasiveness of digital identity systems means that our personally identifying information (PII) is stored in thousands of systems, some of which are inadequately secured. Companies often use this information for purposes beyond what it was initially requested for. The difficulty and expense of running digital identity systems mean that central identity providers can track access requests and correlate people's activities across many sites.

Authorization systems can add to this problem by creating authorization logs that leave a record of access history and require more attributes than are necessary to make a correct access decision. People concerned about their privacy may lie about self-asserted data, making access decisions suspect.

At the same time, dynamic authorization can help with the privacy problem by enforcing privacy protections so that data is accessed only when necessary and by the right parties. Privacy is a critical aspect of digital identity, so we'll discuss it in more detail later in this chapter.

Some approaches to privacy, like *self-sovereign identity* (SSI) (https://www.manning.com/books/self-sovereign-identity), aim to give individuals more control over how their data is shared and used. We'll explore these ideas more deeply in chapter 15, when we discuss decentralized identity and verifiable credentials.

### 2.2.5 The consent problem

Closely related to the privacy problem is the problem of consent. Many identity systems lack the means for people to give granular consent. The lack of control over identity data sharing is often driven by business incentives. The ubiquitous cookie notices that browser users are forced to endure are a shining example of this. Even when they do allow more fine-grained consent, organizations make it so difficult that people are likely to just consent to everything by default.